

DIRECTED ENERGY ENGAGEMENT DIGITAL TWIN

USER GUIDE

PHYSICS-INFORMED • CLOSED-LOOP • SYSTEMS ENGINEERING

MODEL. SIMULATE. VALIDATE. DOMINATE.

Reduced-order engineering analysis. Not a validated operational weapon model.

1. Mission and Scope

The Directed Energy Engagement Digital Twin is a Streamlit-based engineering environment for exploring sensing, tracking, engagement geometry, reduced-order beam propagation, platform power and thermal response, target thermal response, uncertainty, 3-D visualization, verification, reference benchmarking, and export.

MODEL BOUNDARY

Treat outputs as engineering decision-support indicators. The application does not claim validated lethality, probability of kill, operational fire-control authority, or high-fidelity missile aerodynamics.

End-to-End Engagement Loop

1 DETECT	2 IDENTIFY	3 TRACK	4 DECIDE	5 POINT / AIM	6 DELIVER ENERGY	7 ASSESS EFFECT	8 RE-ENGAGE
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Primary capability	What the app provides	Interpretation
Sensor / state estimation	Detection, classification, covariance-based tracking, EKF tools	State knowledge and uncertainty
Aerospace dynamics	3-DOF kinematics and generic 6-DOF rigid-body dynamics	Reduced-order truth-model support
Beam model	Atmospheric transmission, pointing, spot geometry, irradiance	Physics-informed propagation estimate
Power / thermal	Generator + storage limits, cooling, component thermal network	Resource and thermal readiness
Uncertainty	Monte Carlo simulation and distributions	Scenario sensitivity, not certified risk
V&V	Analytical checks, regression checks, NASA GTM reference benchmark, NOAA IGRA atmospheric comparison	Evidence with explicit scope

2. Quick Start

A typical analysis can be completed in seven steps. Configure the scenario first, then execute the engagement and use the specialized tabs to interrogate the result.

Step	Action	What to inspect
1	Configure Scenario	Environment, target, sensors/state estimation, HEL subsystem, platform, 3-D view settings
2	Run Engagement	Top-level engagement state, model-validity assessment, recommendation, readiness score
3	View 3D Digital Twin	Truth trajectory, moving platform, noisy measurement reconstruction, beam/geometry
4	Check Tracking & Estimation	Track quality, covariance, angular and cross-range uncertainty, EKF consistency
5	Review Beam & Thermal Effects	Transmission, spot geometry, average irradiance, target temperature rise, Estimated Thermal Effect Index
6	Run Monte Carlo	Trial count, uncertain inputs, output distributions, sensitivity and robustness
7	Benchmark / Validate / Export	NASA GTM reference benchmark, NOAA IGRA atmospheric-state comparison, JSON/CSV evidence

FASTEST PATH

For a first run, leave defaults unchanged, execute the engagement, inspect the Engagement Loop and 3D Digital Twin tabs, then review Thermal & Power Digital Twin before changing assumptions.

Top-Level Status Interpretation

Indicator	Meaning
Readiness Score	Composite decision-support score derived from modeled sensing, tracking, pointing, power, thermal, and engagement conditions.
Recommendation	ENGAGE / CAUTION / HOLD-style engineering recommendation generated from model state. It is not an autonomous authorization.
Estimated Thermal Effect Index	Normalized engineering indicator of modeled target thermal response. It is not P(effect), Pk, or a validated lethality probability.
Model Validity	Checks whether the selected scenario remains within intended reduced-order assumptions and solver guidance.

3. Scenario Configuration

The left configuration workflow establishes the environmental state, target geometry and motion, sensing assumptions, HEL parameters, platform resources, and 3-D display settings. Change one family of assumptions at a time when performing trade studies.

Group	Typical controls	Engineering effect
Environment	Atmosphere, terrain / visibility, ambient temperature	Changes propagation, sensing, thermal rejection, and validity context
Target	Range, speed, trajectory/profile, initial geometry	Changes LOS rate, dwell, CPA, available engagement time
Sensors / State Estimation	Update rate, measurement uncertainty, covariance assumptions	Changes track quality and pointing knowledge
HEL Subsystem	Optical source power, wall-plug efficiency, optics efficiency, wavelength / beam parameters	Changes electrical demand, target power, spot and irradiance
Platform	Generator power, stored energy, discharge limit, cooling capacity, coolant temperature	Constrains sustained power and thermal readiness
3D Digital Twin View	Display and trajectory visualization controls	Presentation only unless explicitly tied to model settings

AEROSPACE ENVELOPE

The primary directed-energy effect model is intentionally reduced-order. Higher-speed generic aerospace kinematics belong in the Advanced Twin / V&V context and should not be interpreted as validated high-speed target-effect modeling.

Good Trade-Study Practice

- Establish a baseline scenario and export it before changing parameters.
- Change one driver at a time for local sensitivity; use Monte Carlo for coupled uncertainty.
- Record the model version, scenario settings, solver timestep, and evidence source with each comparison.
- Do not convert user-adjustable trade thresholds into formal requirements unless they are baselined and configuration controlled.

4. Engagement Loop and Decision-Support State

The Engagement Loop tab exposes the eight-stage architecture and the principal state used by the decision logic.

1 DETECT	2 IDENTIFY	3 TRACK	4 DECIDE	5 POINT / AIM	6 DELIVER	7 ASSESS	8 RE-ENGAGE
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State	Interpretation
Detection probability	Modeled sensor detection confidence under the selected conditions.
Classification confidence	Modeled identification confidence; not an intelligence-grade identification guarantee.
Track quality	Normalized state-estimation quality derived from covariance behavior.
Cross-range / angular 1-sigma	Uncertainty measures used to characterize track and pointing knowledge.
LOS rate / time to CPA / time to impact	Geometry and kinematic timing quantities.
Aimpoint margin index	Normalized pointing / geometry margin.
Atmospheric transmission	Reduced-order Beer-Lambert transmission estimate.
Power availability ratio	Actual electrical power available relative to requested electrical input.
Target surface temperature / delta-T	Reduced-order thermal-response state.
Estimated Thermal Effect Index	Engineering indicator normalized to an effective failure threshold; not probability of effect.
Thermal / energy margins	Platform resource margins used in readiness logic.

5. State Estimation

The application combines covariance-based tracking metrics with an Engineering Lab EKF. The intent is to expose estimator quality, uncertainty, and consistency rather than hide them behind a single track score.

Core EKF Structure

Prediction: $x_hat(k|k-1) = f(x_hat(k-1|k-1), u(k-1))$
Covariance: $P(k|k-1) = F(k-1) P(k-1|k-1) F(k-1)^T + Q(k-1)$
Gain: $K(k) = P(k|k-1) H(k)^T [H(k) P(k|k-1) H(k)^T + R(k)]^{-1}$
Update: $x_hat(k|k) = x_hat(k|k-1) + K(k) [z(k) - h(x_hat(k|k-1))]$
Joseph covariance update: $P(k|k) = [I-K(k)H(k)] P(k|k-1) [I-K(k)H(k)]^T + K(k)R(k)K(k)^T$

Metric	Use
NEES / NIS	Consistency diagnostics when the required truth / innovation information is available.
Residuals / innovations	Expose measurement-model mismatch and filter behavior.
Track angular 1-sigma	Links estimator uncertainty to pointing knowledge.
Track cross-range 1-sigma	Spatial uncertainty normal to LOS.
Covariance checks	Support positive-semidefinite / numerical sanity checks.

INTERPRETATION

A high track-quality score is not proof that the estimator is statistically consistent. Review covariance, residual, NEES/NIS, and regression evidence together.

6. Atmosphere, Beam Geometry, and Pointing

The beam model is intentionally reduced-order. Atmospheric extinction, diffraction-like divergence, additional divergence, turbulence-induced spread, and pointing uncertainty are combined to estimate target-plane spot geometry and average irradiance.

Implemented Relationships

- Atmospheric transmission: $T_{atm} = \exp(-\tau)$
- Diffraction half-angle: $\theta_{diff} = M^2 \lambda / (\pi w_0)$
- Effective divergence: $\theta_{eff} = \sqrt{\theta_{diff}^2 + \theta_{add}^2 + \theta_{turb}^2}$
- Spot radius: $r_{spot} = \sqrt{w_0^2 + (\theta_{eff} R)^2 + (\theta_{point} R)^2}$
- Average irradiance: $I_{avg} = P_{target} / (\pi r_{spot}^2)$

NOAA IGRA SCOPE

IGRA radiosonde observations constrain atmospheric-state variables such as pressure, temperature, and derived density. They do not empirically validate optical extinction or directed-energy effects.

Quantity	What increases it	What it generally does
Optical depth	Aerosol / Rayleigh / humidity extinction contributions and path length	Reduces transmission
Effective divergence	Diffraction, added optics / aberration term, turbulence spread	Increases spot size
Pointing uncertainty	Track / gimbal / jitter contributions	Increases effective target-plane spot
Range	Longer propagation distance	Magnifies divergence and pointing displacement

7. Power and Thermal Digital Twin

Platform readiness couples electrical supply, optical conversion, waste heat, cooling capacity, stored energy, and component temperatures. The current implementation is a generic reduced-order resource model, not a detailed electrical bus, pump, or NTU heat-exchanger simulation.

Power Balance

$$P_{store,avail} = \min(P_{store,max}, 3600 E_{store} / t_{dwell})$$

Units for the expression above: E_store in kWh, t_dwell in seconds, and power in kW; the factor 3600 converts kWh/s to kW.

$$P_{elec} = \min(P_{elec,req}, P_{gen} + P_{store,avail})$$

$$P_{opt} = \eta_{WP} P_{elec}$$

$$Internal\ heat = conversion\ heat + optical\text{-}train\ heat$$

$$Cooling\ sink = Q_{cool,max} * clamp[(T_{coolant} - T_{amb}) / (T_{limit} - T_{amb}), 0, 1]$$

Component Thermal Network

$$C_i dT_i / dt = Q_i - \sum_j G_{ij}(T_i - T_j) - Q_{i,sink}$$

Node	Role
Laser Source	Principal optical-source thermal node
Beam Director / Optics	Optical-train heat and coupling to coolant loop
Power Electronics	Conversion losses and thermal limit
Energy Store	Stored-energy node with generic storage-loss proxy
Coolant Loop	Collects heat from component nodes
Heat Exchanger	Ambient-limited cooling sink

8. Target Thermal Response and 1-D Fourier Model

The primary target model is an areal lumped-capacitance model. A separate optional 1-D slab solver is provided for transient conduction studies. These are distinct models and should not be conflated.

Primary Lumped Target Model

$C_A \, dT/dt = \alpha I - h_A (T - T_{amb})$
 $\tau = C_A / h_A$
 $I_{thermal} = clamp[\Delta T / \Delta T_{fail,eff}, 0, 1]$

$\Delta T_{fail,eff}$: effective modeled temperature-rise threshold used to normalize the Estimated Thermal Effect Index.

IMPORTANT

The Estimated Thermal Effect Index is an engineering indicator. It is not a probability of kill, probability of effect, or empirically validated damage model.

Optional 1-D Slab

$\rho c_p (\partial T / \partial t) = k (\partial^2 T / \partial x^2)$
Sign convention: positive net heat flux is defined into the slab.
Front net flux: $q_{front}'' = q_{abs}'' - h_f (T_s - T_{inf})$
Back net flux: $q_{back}'' = -h_b (T_b - T_{fluid})$
Explicit stability: $Fo = \alpha \, dt/dx^2 \leq 0.45$
Boundary stability: $dt [\alpha/dx^2 + h/(\rho c_p \, dx)] \leq 0.45$

Model	Use	Not included
Lumped target	Fast engagement-loop thermal response	Spatial conduction, radiation, phase change, temperature-dependent properties
1-D slab	Transient through-thickness conduction study	Full 3-D conduction, thermoelastic deformation, phase change
Component network	Platform thermal readiness	Detailed coolant hydraulics, pump curve, NTU heat exchanger

9. Monte Carlo Uncertainty Analysis

Monte Carlo mode perturbs selected uncertain scenario inputs, runs the engagement model repeatedly, and summarizes output distributions. Use it to explore robustness and sensitivity, not to manufacture a probability claim that the underlying model does not support.

Workflow	Practice
1. Define uncertain inputs	Use physically defensible distributions and bounds.
2. Select trial count	Increase trials until summary statistics are sufficiently stable for the question being asked.
3. Run simulations	Retain scenario and model version metadata.
4. Inspect distributions	Review means, standard deviations, percentiles, tails, and multimodality.
5. Compare scenarios	Use identical uncertainty definitions when comparing designs.

Recommended Outputs

- Readiness Score distribution
- Track quality / pointing uncertainty distributions
- Power and thermal margin distributions
- Estimated Thermal Effect Index distribution
- Sensitivity of outputs to uncertain inputs

STATISTICAL
DISCIPLINE

A Monte Carlo frequency is only as meaningful as the model, input distributions, and event definition beneath it. Do not label the Thermal Effect Index distribution as P(effect).

10. 3D Digital Twin and Model State

The 3D Digital Twin is the primary spatial diagnostic. It should be used to inspect geometry, trajectories, measurement reconstruction, platform motion, and beam relationships alongside the numerical state.

Visual element	Recommended interpretation
Maneuvering Truth - royal blue	Reference / simulated truth trajectory
Moving Platform - cyan	Platform truth or motion state
Noisy Measurement Reconstruction - magenta	Measurement-derived reconstruction; do not confuse with truth
Current Target - blue marker	Current modeled target state
Beam / valid nominal geometry - green	Nominal beam / geometry visualization
Critical states - orange / red	Attention or limiting condition

Model State Tab

Use Model State as the audit-friendly numerical view of the digital twin. It is particularly useful when a 3-D visual appears surprising: verify the underlying position, velocity, uncertainty, geometry, power, and thermal values before drawing conclusions from the graphic.

VISUALIZATION RULE A 3-D rendering is evidence of geometry, not evidence of physical validation. Numerical state, assumptions, and V&V results remain authoritative.

11. Advanced Twin / V&V

Advanced Twin / V&V contains generic aerospace truth-model functions and verification tools that are intentionally separated from the primary directed-energy effect envelope.

Capability	Purpose	Scope
Generic maneuvering truth	Generate controlled aerospace kinematics / attitude behavior	Generic, not vehicle-specific
3-DOF point mass	Reduced-order translational flight dynamics	Simplified aerodynamics
Generic 6-DOF rigid body	Quaternion attitude, body rates, forces and moments	Low-order rigid body; not high-fidelity aircraft aero
Sensor measurements	Generate noisy generic measurements	Synthetic measurement model
Second-order gimbal	Demonstrate generic sensor / gimbal response	Control-model demonstration
Timestep convergence	Check numerical sensitivity to solver step	Verification
Analytical benchmarks	Compare selected calculations against known relationships	Verification
Energy / invariant checks	Check boundedness, conservation-like constraints, quaternion norm	Verification

Reference Frames

App canonical body frame: FLU = x forward, y left, z up. NASA/GTM reference convention: FRD = x forward, y right, z down. Basis transform $S = \text{diag}(1, -1, -1)$. Attitude conversion is performed through the DCM: $C_{\text{new}} = S C_{\text{old}} S^T$.

12. Engineering Lab

Engineering Lab provides the app's strongest systems-engineering layer: measurement-driven EKF analysis, requirements traceability, trade studies, provenance, regression checks, and engineering exports.

Section	Use
Measurement-Driven EKF Tracking	Ingest or generate measurement data, estimate state, evaluate residuals and consistency.
Requirements Traceability	Map needs to system requirements, allocated functions/models, verification cases, metrics, thresholds, evidence, and status.
Scenario / Trade Study	Compare design or scenario alternatives under controlled assumptions.
Model Provenance	Document model scope, evidence source, and validation / benchmarking status.
Internal Regression Checks	Automated checks intended to catch numerical or logical regressions over time.
Engineering Lab Export	Package evidence for downstream review.

INCOSE PRACTICE

Trade-study thresholds are not automatically configuration-controlled requirements. Formal verification should use approved requirement values, a defined baseline/revision, verification cases, and retained evidence.

13. Validation and Reference Benchmarking

The app deliberately separates verification, model-to-model reference benchmarking, and observational validation. Preserve these distinctions when reporting results.

Evidence type	Current app use	Correct claim
Numerical / analytical verification	Convergence, invariants, regression checks, analytical benchmarks	Checks implementation behavior and numerical consistency
NASA GTM reference benchmark	Matched generic 6-DOF flight-dynamics comparison	Model-to-model reference benchmarking; not empirical flight-test validation
NOAA/NCEI IGRA	Radiosonde atmospheric-state comparison	Observational validation/comparison of atmospheric-state variables only
Directed-energy effect model	Reduced-order thermal indicator	No empirical lethality / Pk validation claimed

NOAA Retrieval Behavior

- Current-year requests prefer NOAA's smaller year-to-date archive.
- Downloads are bounded and disk-backed to reduce memory pressure.
- The archive is scanned sounding-by-sounding; only the nearest sounding is retained.
- On retrieval failure, the last valid sounding remains available in session state.

V&V

LANGUAGE

Use 'verified' for numerical/logical checks, 'reference benchmarked' for NASA GTM comparison, and 'observationally compared/validated' only for the atmospheric state supported by IGRA data.

14. Thermal & Power Digital Twin Workflow

This tab supports component-level thermal readiness studies in addition to the engagement-loop platform model.

Control / output	Interpretation
Editable component table	Initial temperature, capacitance, limit, and heat-load allocation for six generic thermal nodes
Simulation duration / output timestep	Controls analysis horizon and reporting cadence; internal timestep is stability-limited
Load multiplier	Scales modeled thermal loading for trade studies
Cooling command	Scales available cooling-sink action
Thermal Readiness	READY / CAUTION / HOLD / INVALID state based on component limits and margins
Limiting Component	Component with the smallest peak thermal margin
Time to Thermal Limit	Observed crossing time, or right-censored if no crossing occurs during the horizon
1-D Fourier model	Optional transient slab-conduction analysis
Frequency-domain diagnostic	Diagnostic for periodic thermal loading; not a replacement for transient physics

RIGHT-CENSORING

If no component reaches its thermal limit during a 120 s run, the correct statement is 'time to limit > 120 s (right-censored)', not that the true limit time equals 120 s.

Thermal Readiness Interpretation

- READY: no modeled component reaches its limit and minimum peak margin remains comfortably positive.
- CAUTION: thermal margin is small even though the modeled limit is not crossed.
- HOLD: at least one modeled component reaches its limit within the simulated horizon.
- INVALID: an initial component temperature is already at or above its modeled limit.

15. Export, Reproducibility, and Reporting

Use the Export and Engineering Lab export functions to retain the numerical state behind screenshots and conclusions. A professional digital-engineering record should preserve configuration, model version, assumptions, outputs, and v&v evidence together.

Export artifact	Recommended contents
JSON	Scenario configuration, key state, readiness, provenance, model/version metadata
CSV	Time histories, Monte Carlo results, requirement evidence, thermal histories
Figures / screenshots	Use for communication; retain the underlying numerical export as the authoritative evidence
Validation / benchmark record	Dataset/source, selected case, metrics, residuals/errors, date, model version

Minimum Report Checklist

- State the scenario and purpose.
- State which model layer produced the result: primary engagement model, Advanced Twin, Engineering Lab, or Thermal & Power Digital Twin.
- State major assumptions and validity limitations.
- Report uncertainty and margins, not only nominal values.
- Use 'Estimated Thermal Effect Index', never 'effect probability' unless a separately validated probability model is introduced.
- Distinguish verification, NASA GTM reference benchmarking, and NOAA atmospheric-state observational comparison.
- Attach or retain machine-readable exports for reproducibility.

16. Key Variables and Glossary

Symbol / term	Meaning
R	Range
LOS	Line of sight
CPA	Closest point of approach
T_atm	Atmospheric transmission
tau	Optical depth
w0	Initial beam radius
theta_eff	Effective angular beam spread
r_spot	Effective target-plane spot radius
I_avg	Average target-plane irradiance
C_A	Target areal heat capacity
h_A	Target areal heat-loss coefficient
deltaT_fail,eff	Effective modeled temperature-rise threshold used to normalize the Estimated Thermal Effect Index
G_ij	Thermal conductance between component nodes
Fo	Fourier number
EKF	Extended Kalman Filter
NEES / NIS	Estimator consistency statistics
GTM	NASA Generic Transport Model reference dataset/model context
IGRA	NOAA Integrated Global Radiosonde Archive
V&V	Verification and validation
Estimated Thermal Effect Index	Normalized engineering thermal-response indicator; not a probability

FINAL MODEL NOTE

This application is a reduced-order aerospace digital-engineering framework intended for modeling, simulation, trade studies, uncertainty analysis, **V&V** , and decision support. Results remain scenario- and assumption-dependent.